
Anti-benching

From the original problem to the causal-split conclusion

A complete account of what we stated, what we changed, what the gates measured, why the attempted cure worsened the locked result, and what the evidence actually permits next

The answer in one paragraph

The original defect was real: the joint pair selector could discard one troll’s available work, leaving that troll idle for as many as 194 consecutive turns. The first cure, P1+P2, removed the visible benching mechanism but usually did not restore progress. Route diagnosis then found useful replant PICK candidates being built and thrown away. Revision r2 preserved those candidates, but also introduced persistent cross-turn commitments and a duplicate-candidate path. On the frozen 240-game cost panel, blocking rose from **35 to 115**; five orchard-inertness failures alone were decisive. That result is valid and r2 is rejected. The later review corrected one overstatement: the data do *not* prove that persistent commitment caused all or most of the damage, and 73 P4 labels are not 73 directly caused stall windows. The clean conclusion is: **the option is structurally isolatable, but its value is unproved**. A future experiment, if authorized, must add only the ephemeral current-turn option to the exact baseline candidate list, without duplicate candidates or new commitment memory, and must stop at the first structural or progress falsifier.

Prepared by

local_codex_1

Scientific verdict

RESULT_VALID_BUT_CAUSAL_CLAIM_UNPROVEN

Canonical state

2026-08-24 · artifact line through bc68b712

Engineering verdict

r2 rejected; Arena closed

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1 Executive decision

This report covers one continuous investigation. It begins with the anti-benching rule, follows the first selector repair and the route diagnosis, reconstructs r2 and its gates, and ends with the review-corrected causal conclusion and the exact boundary of a possible next experiment.

Operational decision. Revision r2 is **STOP**. It produced 115 blocking games against 35 for the exact P1+P2 base on the same locked 240-game panel: 80 de-novo blocks, zero healed blocks, five new P3 failures, and 73 new P4 labels. G-e progress testing was correctly not run after the first hard G-d falsifier. Nothing in the later review reopens r2.

Scientific correction. The measured result is reproducible, but the broad causal story was stronger than the evidence. R2 bundled at least three mechanisms. There was no no-commitment arm, no Delta-B census over the full panel, and no per-game selector decomposition. Moreover, P4 uses future trajectory information, so some labels describe temporal reclassification rather than a stall caused inside the reported earlier interval. The precise verdict is therefore **RESULT_VALID_BUT_CAUSAL_CLAIM_UNPROVEN**.

	What is settled	What remains open
Problem	Benching exists; the selector can discard available work.	How much economically useful work is recoverable.
P1+P2	Mechanism changed; visible benching fell.	Why most newly active turns failed to progress.
r2	Locked cost result worsened decisively; reject it.	Which bundled mechanism caused which changed game.
Delta-A option	Naturally forms and can survive selection; structurally isolatable.	Durable progress, score, full-corpus reach, and closed-loop safety.
Next action	Design contract and falsifiers are written.	Implementation and measurement require a new authorization.

2 What we stated initially

2.1 The rule and the failure shape

The original statement was not “make both units move.” It was stronger and more useful:

Anti-benching rule R-2. When a troll has oracle-verified available work, the team’s joint pair selection must not discard that work and leave the troll idle. Employment means real progress, not merely replacing WAIT with a harmless action.

The target was the 24-case **GOAL_SPLIT** class. The recurring shape was that each local route could be individually reasonable while their joint pairing was not. One unit wanted to move into the cell occupied by its teammate, while the selected partner command ordered that teammate to WAIT; the referee then dropped the move. In the worst observed case this repeated for **194 turns in a 200-turn game**.

The field audit showed the defect was sparse but real: across 160 real games, trolls had work available but received nothing on **615 of 84,928 troll-turns (0.72%)**. That audit motivated the work; it did not imply that every later fixture turn or every restored candidate was one of those 615 turns.

2.2 Initial success criterion

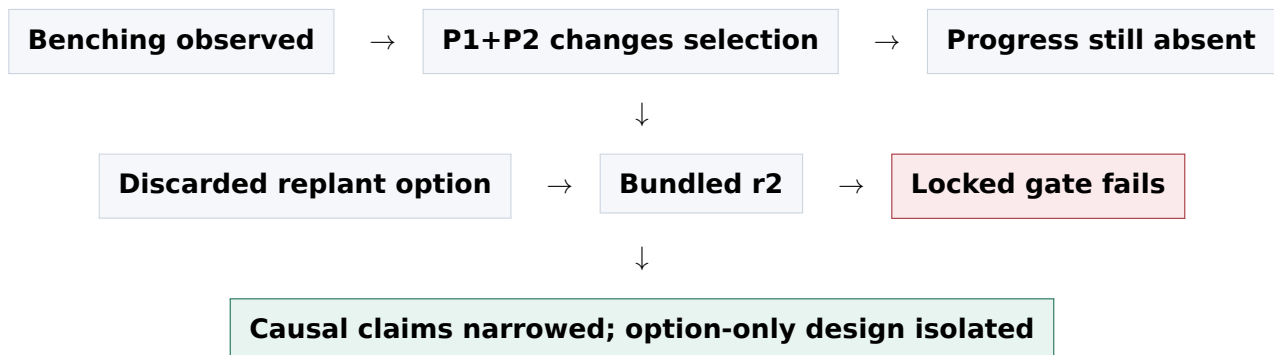
The investigation deliberately separated three questions:

1. **Mechanism:** does the selector stop constructing self-impossible or needlessly idle pairs?
2. **Progress:** does the previously benched troll actually bank, harvest, build, route, or otherwise advance useful work?
3. **Cost:** does the cure preserve named invariants and avoid new liveness failures on the locked panel?

This separation became decisive. A command can silence the benching detector without helping the game. Conversely, a later action can alter a trajectory-level liveness label without causing the earlier interval that the label reports.

3 The investigation as a sequence

Phase	Question	Work performed	Result
1 / WHY	Why is work discarded?	Instrumented the real selector arithmetic and hard filters; named pair conflicts and exact ties.	Mechanism localized.
2 / P1+P2	Can a minimal selector repair employ the troll?	Refused self-impossible pairs; broke exact ties toward fewer waits; tested two exact bases.	Benching fell, progress mostly did not.
3a / route	What does the newly active troll do?	Traced candidate formation and returns for OSC-004/013/017; inspected the lossy fallback.	Replant PICKs were formed then discarded.
3b / r2	Can preserving those PICKs restore progress?	Built an extend-not-replace revision with persistent regeneration behavior; independently reviewed reach and build identity.	Natural reach shown; value not yet shown.
G-d	Is the candidate safe on named costs?	Ran and independently reproduced the complete locked 240-game panel.	35 → 115; hard stop.
Rereview	Is the causal explanation sound?	Fresh review separated observation, deduction, hypothesis, and missing counterfactuals.	Result valid; causal claim unproved.
Causal split	Is a cleaner mechanism test possible?	Read-only source/evidence analysis wrote an exact option-only contract.	ISOLATABLE; still unvalidated.



4 Phase 1 and Phase 2: the first cure

4.1 What P1 and P2 changed

The first repair was intentionally narrow and applied as one patch to two bases.

Part	Rule	Intended effect
P1	Refuse a pair where a unit moves onto a cell occupied by a teammate whom the same pair orders to WAIT.	Remove self-impossible pairs that the referee will invalidate.
P2	On an exact pair-score tie, prefer the pair with fewer WAIT commands.	Avoid arbitrary map-order tie resolution when an equally scored active pair exists.

P3 and P4 were explicitly not part of that build. The 235 non-deadlock turns identified by the diagnosis remained outside the initial fix. The patch was behavior-changing, so qualification required not just local examples but a named-cost gate and a separate platform decision.

4.2 The repair was mechanically real

The cure did what its code said. On the selected fixtures, benched turns fell to zero. On broader panel accounting, blocking totals also fell:

Base	Before P1+P2	After P1+P2	Change
cure-C	53	33	−20
door-1	43	35	−8

That is why it would be wrong to say the first cure “did nothing.” It changed the intended selection mechanism. The failure was at the next semantic layer: activity was not the same as productive progress.

4.3 Why detector success was not cure success

Under the fixed progress grader, cure-C moved only **3 to 4** and door-1 remained **8 to 8**. Only OSC-034 became fixed on cure-C. OSC-004, OSC-013, and OSC-017 became detector-quiet while retaining `progress_restored = false`.

Two additional named costs appeared on door-1:

- **P3 absolute regression:** m004, seat 0, violated the locked orchard-inertness comparison.
- **P4 and horizon cost:** m021 acquired a new P4 / r5-horizon property violation, even though that game was already blocked for another reason.

First central lesson. “The troll is no longer benched” is an intermediate mechanism metric. It is not the project outcome. A cure must produce actual banking or employment and then survive the named-cost gate.

5 Phase 3a: route diagnosis

5.1 The lossy fallback

The route probe found that the parent candidate list was not genuinely empty. It typically had a seeded WAIT, and in eligible late states the regeneration path formed valuable-looking replant PICK candidates. But the fallback logic behaved conceptually as follows:

```
build ordinary candidates into out
form safe-regeneration PICK candidates in out
if idle_regeneration and chop candidates are empty:
    return fresh [WAIT]          # earlier out is discarded
```

The open design question was therefore **extend versus replace**: should the fallback return a fresh wait-only list, or extend the exact parent fallback with the candidates that had just been formed?

5.2 OSC-013: the clearest trace

Across 170 idle turns in OSC-013, the route split cleanly:

Turns	Count	Observed route state
31-99	69	Only WAIT; there was no discarded replant option to recover.
100-200	101	Two replant PICK candidates were present in out, with scores 7500 and 7499, but the fallback returned WAIT.

This was strong evidence of a missing option. It was not yet evidence that selecting the option would create durable progress. It also did not show that a persistent intention to complete the replant transaction was safe.

6 Phase 3b: what r2 actually bundled

Revision r2 implemented the intuitive “extend, do not replace” idea, but its behavioral surface was larger than that phrase suggests.

Mechanism	What it did	Status before G-d
Delta-A	Preserved the replant PICK candidates that the parent fallback discarded.	Natural formation and selection observed.
Persistent commitment	When a new PICK was selected, recorded regeneration state and routed the unit toward endgame candidates on later turns.	Source path identified; no no-commitment control.
Delta-B	In carried, shack-adjacent fallback states, returning the whole accumulated vector and appending bank candidates could produce element-identical duplicates.	Derived from source; zero fixture ticks; unmeasured on full panel.
Joint selector exposure	The unchanged pair selector saw a larger candidate cross-product and could choose a different command for either unit.	Selection occurred; alternatives not decomposed per game.

The source structure matters. Candidate formation occurs in `main_candidates`; joint choice occurs in `MoisanBot::select`; cross-turn memory is inserted by `remember_selected_regeneration`; later command routing reads that memory. Thus a small candidate-list edit can propagate into later turns through a separate memory mechanism.

Why the bundle matters. The `r2` experiment did not compare “baseline” with “the exact same bot plus one ephemeral option.” It compared the baseline with a system in which the option could be selected, remembered, and followed across turns, while another duplicate-candidate path also existed. The experiment can reject that system. It cannot identify every component’s causal share without additional arms or traces.

7 Pre-panel evidence: formation, selection, and reach

The build and reach reviews established that Delta-A was not merely a synthetic fixture idea.

7.1 Fixture library

Base	Effect	No effect	Formed	Selected	Delta-B
cure-C	20	14	203	144	0 ticks
door-1	19	15	201	143	0 ticks

All first selected turns were turn 100, matching the late-game precondition. Zero Delta-B fixture ticks means “not observed in this fixture library,” not “impossible.”

7.2 Real-corpus natural reach

Only **49 of 160** real games could be parity-verified; **111 refused closed**. Within the verified subset, the analysis found:

Eligible NOTHING/NOTHING rows	882
Restored-and-selected turns	339
Episodes	34
Games containing those episodes	14
Changed whole command-vector turns	255

These numbers prove exact natural reach in the accepted subset. They do *not* mean “339 of the original 615 benched turns were cured,” and they must not be divided by an unrelated 2,903-row denominator. They also do not establish full-corpus representativeness, progress, value, score, or safety.

8 The locked G-d result

8.1 The decisive measurement

The full named-cost panel contained 240 locked games. Against the exact door-1 P1+P2 base, `r2` produced:

Measure	Exact base	r2
Blocking games	35	115
Healed games	0	
De-novo blocking games	80	
New P3 orchard-inertness failures	5	
New P4 liveness labels	73	
New horizon failures	0	

The independent local reproduction matched all 240 per-game rows and the exact 85-game changed decomposition. chatgpt_1's fresh-eyes review also returned BLOCKED. This is not a sampling wobble or an analyzer disagreement.

8.2 The five direct P3 failures

P3 required complete command-stream equality on orchard-eligible initial layouts. Five games diverged directly at turn 100:

Game / seat	First turn	r2 command	Exact-base command	Frozen label
m035 / 0	100	WAIT; PICK 2 BANANA	WAIT; WAIT	P3, P4
m065 / 0	100	PICK 0 BANANA; PICK 2 PLUM	WAIT; WAIT	P3, P4
m074 / 0	100	PICK 0 BANANA; WAIT	WAIT; WAIT	P3
m104 / 0	100	WAIT; PICK 2 APPLE	WAIT; WAIT	P3
m114 / 0	100	PICK 0 BANANA; PICK 2 PLUM	WAIT; WAIT	P3, P4

One P3 divergence was enough to fail the frozen gate. Five were observed. Therefore G-e—the later progress gate—was correctly left unrun. A positive progress number could not waive the hard cost failure, and running it would have weakened the precommitted first-falsifier discipline.

Why the result worsened. In the exact operational sense, r2 broadened the behavior of the bot in states covered by the locked safety rules. It changed commands in five maps that had to be inert and generated many trajectories classified as newly blocking. Since it healed no blocked game, the net locked result moved from 35 to 115. This is sufficient to reject r2 even before we know which bundled internal mechanism caused each change.

9 Why the first causal explanation was too strong

The early explanation was plausible: the newly selected PICK creates a persistent regeneration commitment; that commitment keeps routing the troll and causes long liveness regressions. Source inspection supports the possibility. The panel supports broad downstream behavioral change. But plausibility plus correlation is not a component-level causal estimate.

9.1 Missing counterfactuals

The experiment did not include:

- an r2 arm with Delta-A but no new commitment memory;
- a full-panel census and same-state selector fork for Delta-B;
- per-changed-game candidate vectors, rejected pair alternatives, and scores;
- commitment timelines joined to first divergence and later progress events; or
- a measurement that partitions P4 windows by whether they occur before, cross, or after the first command divergence.

Without those counterfactuals, “persistent commitment caused most of the 73” remains a hypothesis.

9.2 P4 looks backward from the future

The frozen P4 implementation computes remaining work and a live horizon from the later trajectory, then scans backward and trims windows relative to the terminal suffix. That definition is valid as a gate, but it is future-dependent.

The smallest concrete counterexample is m035, seat 0:

Candidate-only P4 win- turns 33-99
dow

live_end 99

terminal_from 106

First command diver- turn 100
gence

Commands and world identical between exact base and r2
through turn 99

A deterministic referee cannot make a command at turn 100 cause different commands or world state on turns 33-99. Instead, the later reactivation changes where the final terminal suffix begins, which retroactively changes the earlier interval’s P4 classification. The same temporal shape is seen in m065 and m114.

Correct interpretation. “73 new P4 labels” is a valid frozen-gate output. It is not a causal inventory of 73 stalls that r2 produced inside the named intervals. Some subset may still be genuine post-divergence harm; the existing package does not partition it.

This correction does not rescue r2. The five direct P3 command divergences remain, and the exact 35-to-115 gate result remains. It only prevents the project from learning the wrong mechanism from a correctly failed candidate.

10 Causal ledger

The final review separated four epistemic levels: observed, deduced from source or definitions, hypothesized, and missing evidence.

10.1 Delta-A: preserved replant option

Observed. Two score-7500/7499 candidates were discarded on 101 consecutive OSC-013 turns; the fixture library reproduced formation and selection; the verified real subset reproduced 339 restored-and-selected turns. The five P3 games show turn-100 command impact.

Deduced. The option has non-zero natural reach and can survive joint selection.

Hypothesis. An isolated one-turn option may create useful banking or employment at an acceptable cost.

Missing. The progress gate, closed-loop value, full-corpus reach, score, and qualification.

10.2 Persistent regeneration commitment

Observed. The tuned r2 path records selected PICKs and later routes remembered units; source separates memory insertion from formation and selection.

Deduced. A selected Delta-A action can have a larger cross-turn behavioral surface than its local candidate diff suggests. The new option can be excluded from memory without changing baseline PICK handling.

Hypothesis. Persistent commitment caused most post-turn-100 regressions.

Missing. A no-commitment arm and field-level memory timelines.

10.3 Delta-B: duplicate bank candidates

Observed. Source derivation shows duplicated, element-identical bank candidates in a carried, shack-adjacent fallback state. The fixture library reached zero such ticks.

Deduced. Delta-B and Delta-A are mutually exclusive in one unit state ($\text{carried} > 0$ versus $\text{carried} == 0$), but a Delta-A trajectory could reach Delta-B later.

Hypothesis. Exact duplicates are selector-inert—or they alter an ordering or tie path.

Missing. A natural-state census and same-state selection comparison on the 240-game run.

10.4 Joint pair selection

Observed. The selector applies target, stock, self-blocking, score, and fewer-wait checks across the candidate cross-product. Added candidates were sometimes selected.

Deduced. Adding one unit option can change either selected command without changing the selector's source.

Hypothesis. Some options complement a partner command; others displace a better pair.

Missing. Per-game full vectors, pair scores, rejected alternatives, and resolved commands.

10.5 P3 and P4

P3 observed and decisive. Five orchard-eligible streams differ from the exact parent. This is direct, not inferential.

P4 observed but causally ambiguous. Seventy-three labels are reproduced exactly under the frozen definition. At least one named window precedes the first command divergence, so the total is not a direct stall count.

11 The final causal-split conclusion

Conclusion: ISOLATABLE

The missing replant option can be separated structurally from both new persistent commitment and duplicate bank candidates. Candidate formation, joint selection, and commitment insertion occupy distinct functions. A future design can construct the exact parent's returned list first, append only the explicitly discarded replant candidates, pass the result through the unchanged selector, and exclude any selected added candidate from commitment memory.

This is an isolatable design, not a validated cure.

11.1 Exact option-only candidate-list contract

Let $B[u]$ be the exact ordered candidate vector returned by the parent for unit u on the same recorded decision state. Let $D[u]$ contain only the eligible, deduplicated replant PICK candidates. The candidate vector under test must be:

$$C[u] = \begin{cases} B[u] \parallel D[u], & \text{when every exact precondition holds,} \\ B[u], & \text{otherwise,} \end{cases}$$

where $\parallel$ means ordered append. No element of B may be removed, reordered, rescored, retargeted, or duplicated. Nothing other than D may be added.

11.2 Exact same-state preconditions

Delta-A may form for a unit only when all conditions hold:

1. command routing uses the parent's ordinary `main_candidates` branch—not an early, endgame, or remembered-regeneration branch;
2. the dedicated option-only flag is enabled and the frozen initial layout is not orchard-eligible;
3. idle regeneration is active and chop candidates are empty;
4. carried inventory is zero and free capacity is positive;
5. turn is at least 100, there are at most two plants, and at least two own units exist;
6. the unit is adjacent to its own shack and no plant occupies its cell; and
7. the bank contains at least one fruit kind.

For each bank fruit in unchanged inventory order, D uses the existing identity: `PICK <unit> <kind>`, score 7500 – priority, target the unit's current cell. Any element already identical in B is removed from D .

11.3 What must remain exact

Parent candidate list	Byte/bit exact and ordered before the append.
Selector	Source-identical: compatibility, stock checks, self-blocking, scores, wait count, unit order, and conflict resolution unchanged.
Sibling unit	Exact parent vector unless it independently satisfies all Delta-A conditions.
Delta-B	Absent structurally; never return the failed r2 arm's whole accumulated out.
Memory	Selected <i>D</i> creates, renews, and changes no persistent field. Parent-existing PICKs retain exact behavior.
Provenance	Current-turn only, cleared before the next turn, and invisible to scoring or ordering.
Orchard maps	<i>D</i> empty on every turn; complete streams byte-identical to the parent.

Later turns may differ because the referee executed a selected PICK and the world now holds different cargo. They may not differ because the experimental candidate inserted a hidden target, phase, fruit, owner, or commitment.

12 What a future one-hour stage should do

No future experiment was authorized by this report. If the project opens one, the order below is the minimum useful path. Each row has an early falsifier so failure stays cheap and interpretable.

Step	Claim	Smallest measurement	Stop immediately if
1	Only the contracted list delta exists.	Source-diff allowlist and same-state generator fork recording complete ordered <i>B</i> , <i>D</i> , and <i>C</i> vectors.	Any change outside <i>D</i> ; parent removal/reordering; duplicate; unrelated sibling change.
2	Delta-A creates no commitment.	Snapshot every persistent field before/after selection and on the next turn; include a deliberately committing poison control.	Selected <i>D</i> changes any persistent state, or baseline PICK handling changes.
3	Selector remains exact and can select <i>D</i> .	Replay the unchanged selector on identical states with <i>B</i> and <i>C</i> ; record scores, compatibility, selected pair, and resolved commands.	Selector hash changes, illegal pair appears, or <i>D</i> never survives selection.
4	Orchard-inert scope is exact.	Predicate parity on boundary fixtures plus complete stream equality on every orchard-eligible locked game.	One predicate disagreement or one command byte differs.
5	The isolated option restores real progress.	Existing two-clause progress grade on a matched benched-and-work-available population, with selected-option provenance.	No positive population improvement, detector-only silence, or a required fixture lacks progress.

Step	Claim	Smallest measurement	Stop immediately if
6	Closed-loop costs are acceptable.	Only after progress: full locked named-cost panel under a newly frozen protocol.	Any P3 divergence, new P4/horizon event under that protocol, worse blocking, or unnamed changed game.
7	P4 labels can be interpreted causally.	Separate read-only evidence tool joining first divergence, progress windows, work remaining, and live horizon for the 73 existing rows.	It cannot reproduce the frozen labels or calls a pre-divergence interval command-caused.
8	Local value transfers to score.	Only after qualification: separately authorized paired score design with balanced arm order.	Candidate is not qualified or identities/order are ambiguous.

12.1 Stage-level stop rule

If the isolated option has natural selection but no useful progress, retire the replant cure family and move to explicit capability-aware work/resource ownership. If the option is useful but requires memory to complete a transaction, design that memory later as a bounded protocol with explicit success, timeout, cancellation, and ownership. Do not restore the generic commitment implicitly.

P4 semantics are a separate evidence-tool question. Clarifying them may improve future causal language, but it cannot waive r2's P3 failures or rewrite the 35-to-115 result.

13 Final conclusions

PROVED	The original benching defect and its 194-turn worst case were real.
PROVED	P1+P2 changed the intended selector mechanism and reduced visible blocking.
PROVED	Activity alone did not restore progress in most target fixtures.
PROVED	The parent route discarded naturally forming replant options.
PROVED	R2 reached natural states and changed commands.
PROVED	The frozen G-d result worsened from 35 to 115 and was reproduced exactly.
PROVED	Five direct P3 divergences independently reject r2.
PROVED	Delta-A can be isolated in design from new commitment and Delta-B.
NOT PROVED	That persistent commitment caused most of the 73 P4 labels.
NOT PROVED	That 73 P4 labels equal 73 stalls caused inside their reported windows.
NOT PROVED	That Delta-B was inert on the full panel.
NOT PROVED	That the isolated option restores durable progress.
NOT PROVED	Full-corpus representativeness, score benefit, qualification, or Arena readiness.

The project-level conclusion. We found a real planner defect, applied a real but semantically insufficient first repair, then built a broader revision whose locked safety result was decisively worse. Independent reproduction and review confirmed the rejection. The rereview did not overturn the result; it improved the science by narrowing the causal statement. The next profitable move is therefore not to retry r2 or debate the frozen count. It is to test—if separately authorized—the smallest option-only mechanism whose candidate-list delta, memory behavior, selector identity, and orchard inertness can all be falsified before a costly panel run.

No source, bot, detector, grader, panel, replay corpus, simulation, TestSession, submission, resident, or Arena state was changed or run in producing this report.

A Evidence map and immutable pins

The narrative above is a synthesis of the following canonical artifacts. Branch-plus-commit pins are included where the evidence is not on the final integrated line.

Purpose	Canonical evidence
Original task and rulings	coordination/tasks/20260820-pair-selector-anti-benching.md
Original project statement	docs/GOALS.md docs/RULES-LEDGER.md docs/DISCOVERY-two-correct-doors-make-a-wall-2026-08-17.md
Phase 3a route diagnosis	agent/claude_1@09ed550f91936818425ad2611c1b875531f32a35:
r2 design	claude_1/picker2/phase3-generator-route-2026-08-20.md agent/claude_1@09ed550f91936818425ad2611c1b875531f32a35:
Exact base and r2 source	claude_1/picker3/phase3b-design-proposal-r2-2026-08-22.md agent/claude_1@e6cb7523d87d4da02e6f81406d572e3e83e4cf10:
Build and reach review	claude_1/picker2/candidate-door1-p1p2.rs claude_1/picker3/candidate-door1-p3b.rs agent/codex_1@35d569f2b78c90dd7c15b46183376cc95efa7196:
Locked G-d result	codex_1/reviews/pair-selector-phase3b-build-review-2026-08-23.md codex_1/reviews/pair-selector-phase3b-reach-review-2026-08-23.md agent/codex_1@35d569f2b78c90dd7c15b46183376cc95efa7196:
Executable reproduction	codex_1/picker3/results/gd-door1-panel-2026-08-23.md codex_1/picker3/results/gd-door1-decomposition-2026-08-23.json agent/local_codex_1@16b6e4ada72ab1381833162ed98e97ba930cd9b4:
Fresh causal rereview	local_codex_1/reviews/pair-selector-gd-ge-unified-review-2026-08-23.md agent/chatgpt_1@a3d2b02a605800d147cc78b9995a7a3525b9e315:
Final causal split	chatgpt_1/reviews/anti-benching-result-strategy-rereview-2026-08-23.md local_codex_1/reviews/anti-benching-causal-split-design-2026-08-24.md;
Final integrated line	artifact c51f8260fd90cde20193ad4ded38e7b1290ca202 bc68b7120e7ff51cc60f1a04efe7637885ec83cc

The sacred resident source remained byte-exact at SHA-256:

```
fff6669b0bc0b15b0992637f70c07197e1838f403cb7fd038bc1fae73d52b13f
```

It remained exact throughout the final review and report work.

B Terminology

Benched turn	A troll has oracle-verified available work but receives no effective work from the joint command.
Detector-quiet	The detector no longer emits the original symptom; this does not imply useful progress.
Delta-A	Only the replant PICK candidates formed by the safe-regeneration block and discarded by the parent idle fallback.
Delta-B	Element-identical bank candidates introduced by r2's whole-vector return plus another append path.
P3	Frozen orchard-inertness rule: the candidate's complete command stream must equal the exact parent on eligible initial orchard layouts.
P4	Frozen trajectory-level long-stall classifier using remaining work and a future-derived live horizon.
G-d	Full locked named-cost panel; the first decisive safety gate for r2.
G-e	Progress gate scheduled after G-d; correctly unrun because G-d failed.
Exact base	The pinned door-1 P1+P2 source and runtime identity used by the accepted package.
Isolatable	Source structure permits a clean component experiment. It does not mean useful, safe, qualified, or authorized.

C Chronology

2026-08-20	Owner charts anti-benching under rule R-2; Phase 1 mechanism probe begins; P1+P2 design approved.
2026-08-20/21	Dual-base P1+P2 build and gates show the selector mechanism changes, but progress mostly does not recover; P3/P4 costs appear.
2026-08-21	Owner chooses "revise"; Phase 3a diagnosis is required before another build.
2026-08-20-22	Route analysis isolates the lossy regeneration fallback; r2 design is proposed and built.
2026-08-23	Independent build/reach review accepts natural Delta-A reach but leaves Delta-B unmeasured; locked G-d panel reports 35 to 115.
2026-08-23	local_codex_1 reproduces all 240 rows; chatgpt_1 independently returns BLOCKED; G-e remains unrun.
2026-08-23/24	Fresh rereview corrects the broad causal story; result remains valid and candidate remains rejected.
2026-08-24	Causal-split memo concludes ISOLATABLE and writes the exact option-only contract and falsifier-ordered future matrix.
2026-08-24	This complete narrative report consolidates the investigation without opening a new experiment.

Document source: local_codex_1/reports/anti-benching-complete-story-2026-08-24.tex.

Rendered artifact: local_codex_1/reports/anti-benching-complete-story-2026-08-24.pdf.